

Dynamic Portfolio Allocation in Goals-Based Wealth Management *

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Abstract

Given any set of exogenously provided efficient portfolios, we develop a dynamic programming algorithm that constructs an optimal portfolio trading strategy to maximize the probability of attaining an investor's specified goal wealth at the end of a designated timeframe. Our algorithm can also accommodate periodic infusions or withdrawals of any size with no degradation in runtime performance. We explore how the terminal wealth distribution is sensitive to restrictions on the segment of the portfolio's efficient frontier made available to the investor. Because our algorithm's optimal strategy is on the efficient frontier, allowed to depend on the investor's wealth, and allowed to depend on the investor's individual goals and specifications, we show that it soundly beats the performance of target date funds for attaining investors' goals. These optimal goals-based wealth management strategies are useful for modern day FinTech offerings, both advisor-driven or robo-driven.

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1 Introduction

Goals-Based Wealth Management (GBWM) refers to the management of an investor's portfolios with a view to meeting long-term financial goals, as opposed to only optimizing a risk-return tradeoff. In this paper, we present and analyze a dynamic programming approach for GBWM that is fully optimal in meeting long-term goals, while also optimizing risk-return tradeoff. We will show that our approach has several advantageous features in terms of speed and adaptability, and that it soundly outperforms portfolios based on current widespread approaches, such as the target-date paradigm.

GBWM is a modern implementation of behavioral ideas from early work by [Shefrin and Statman \(2000\)](#) espousing behavioral portfolio theory that were further discussed by [Nevins \(2004\)](#) from a practitioner viewpoint. A specific framework for GBWM was initially proposed by [Chhabra \(2005\)](#) and is now followed by many practitioners (see [Brunel \(2015\)](#)). Our approach in this paper is also cognizant of prospect theory ([Kahneman and Tversky \(1979\)](#))¹ and mental accounting ([Thaler \(1985\)](#); [Thaler \(1999\)](#)),² as well as paradigms for how investors form both upside and downside goals, such as aspirational goal-setting ([Lopes \(1987\)](#)), a safety-first criterion ([Roy \(1952\)](#)), and loss aversion ([Shefrin and Statman \(1985\)](#)).

We combine these seminal GBWM ideas for achieving long term goals with modern portfolio theory ([Markowitz \(1952\)](#)) to achieve the short term goal of minimizing the risk associated to any specific expected return. GBWM is fully consistent with Markowitz's mean-variance theory. This was developed in [Das, Markowitz, Scheid, and Statman \(2010\)](#) as a static model for mental accounts, while a full analysis of the static model in a GBWM context is provided in [Das, Ostrov, Radhakrishnan, and Srivataav \(2018\)](#).

These static models are not fully optimal, because they can only be implemented period by period in a myopic manner. Previous fully optimal, dynamic approaches to GBWM include solving continuous-time partial differential equations, as shown in the early work by [Merton \(1969\)](#); [Merton \(1971\)](#), or using continuous-time martingale methods based on seminal ideas in [Cox and Huang \(1989\)](#), as shown in more recent work by [Wang, Suri, Laster, Almadi \(2011\)](#) and [Deguest, Martellini, Milhau, Suri, and Wang \(2015\)](#). In this paper we introduce a discrete time fully optimal dynamic programming approach that has many advantages, including being adaptable to many

¹Prospect theory is aimed at modeling how people realistically make decisions, rather than an optimization framework, which usually does not account for all the criteria used by people when making tradeoffs between gains and losses.

²Mental accounting is the paradigm where people behave as if they have different risk-return preferences depending on the goal being achieved, being risk averse in some settings and even risk seeking in others. In our context, mental accounting is the idea that financial goals may be sharply defined and better managed in separate portfolios.

financial situations, being simple to implement, and being fast to run. For example, our approach handles periodic infusions and withdrawals of varying amounts, as well as bankruptcy, which is not always easy to do with the aforementioned continuous-time approaches. Also, our approach employs a polynomial time algorithm that corresponds in general to a runtime of only a few seconds. These features give our algorithm wide applicability, enabling either traditional financial advisors or robo-advisors to quickly and optimally determine the best strategy for investors to pursue to meet their individual wealth goals.

The essence of GBWM lies in choosing a different objective function for dynamic optimization than that chosen in utility-based portfolio problems. The simplest form of the GBWM objective function is probabilistic: that is, define a portfolio goal wealth G at the defined horizon T and find the dynamic portfolio strategy that maximizes the probability of achieving the goal, i.e.,

$$\max_{\{A(0), A(1), \dots, A(T-1)\}} Pr[W(T) \geq G],$$

where $W(T)$ is the terminal portfolio wealth value and $A(t)$ are the possible allocations among the funds available to the investor at each time $t = 1, 2, \dots, T - 1$. The optimal allocation at any time t is a function of $W(t)$, the level of wealth at time t ; G , the goal wealth; T , the timeframe; and other investor specific parameters.

This approach may be contrasted with a utility-based approach where the dynamic strategy is chosen to maximize the expected utility from any consumption over time, as well as the expected utility of the final wealth. This approach, as well as GBWM, are attractive since they focus on long-run outcomes and are not myopic one-period optimization models. Single-period optimization changes the focus from achieving long-term goals to short-term risk-return tradeoffs.

Hybrid approaches, where the objective function to be maximized is expected utility subject to goals as constraints, have also been attempted as in [Deguest, Martellini, Milhau, Suri, and Wang \(2015\)](#). However, this approach requires choosing a utility function for the investor which is hard to determine, and therefore an ad-hoc choice is often required. In this paper, we do not require a utility function to be specified. We provide an optimal solution procedure to the GBWM problem and examine the properties of the trading strategy dictated by the algorithm.

The following briefly characterizes our solution: (i) The backward recursion algorithm that solves the dynamic problem is between quadratic and cubic in its dependence on the timeframe T , quadratic in the granularity of the wealth grid, and linear in the number of portfolio choices. In our base case timeframe of ten years, adding 50 more wealth nodes at each time, and allowing for 15 portfolio choices from the efficient frontier, our algorithm runs in under five seconds. As the markets move up or down through the investment tenure, we rebalance the portfolio every year

such that the probability of reaching the goal wealth continues to remain the highest. (ii) The solution scheme is quite robust to the granularity of the wealth grid and the number of portfolio choices, which suggests that expanding the scale of the problem with additional features, such as tax optimization, can be performed without significantly degrading run times or the solution's accuracy. (iii) The optimal allocation is intuitive in the sense that when the portfolio is far from its goal due to underperformance, risk is increased in order to enhance the probability of reaching the goal, and when the portfolio is outperforming, risk is dialed back to reduce the risk of missing the goal. Therefore, the strategy depends on both time and state (wealth), unlike target-date fund strategies, which only depend on time and cannot accommodate investor-specific goals. (iv) The dynamic program uses a collection of exogenously provided efficient portfolios, so an asset management team can develop the set of model portfolios, while, acting separately, the optimization team can tune the dynamic programming algorithm. This offers a plug-and-play approach to GBWM, where portfolio construction is separated from portfolio allocation.

Our GBWM strategy allows for a variety of important features and results: (i) The final wealth distribution may be modulated by limiting the range of available portfolios (i.e., controls) on the efficient frontier, which exogenously alters the risk that is taken. Curtailing choices at the lower end of the efficient frontier, i.e., raising the minimum risk available to the investor, increases the right tail of the wealth distribution more than the left tail. Conversely, raising the cap on the upper end of the efficient frontier, i.e., raising the maximum risk available to the investor, increases the left tail more than the right tail. This suggests that investors in a GBWM environment who wish to have higher returns are better off raising their minimum risk, instead of the more intuitive move of raising their maximum risk. That said, raising the minimum risk will decrease the chance of attaining the investor's goal wealth, while raising the maximum risk will increase this chance. (ii) The algorithm is flexible: (a) If desired, the algorithm can optimize for multiple wealth goals at the end of the portfolio horizon instead of a single wealth goal, with weights for the relative importance of these multiple goals that the investor can decide. (b) The algorithm can include an investor's specified portfolio infusions and withdrawals. We will see that even small infusions can increase the probability of reaching the goal wealth substantively. When there are withdrawals, there is a chance that the investor will run out money (i.e., go bankrupt) during the investment timeframe. Our algorithm can determine this chance of bankruptcy, and much more importantly, show how to minimize it. (c) The algorithm can accommodate any desired time period between rebalancing and between infusions or withdrawals. (iii) Our algorithm allows us to determine how to optimize retirement savings. For example, our algorithm will allow us to explore the effect of infusions on the minimized probability of going bankrupt in retirement. In particular, we will consider a 50 year old investor who currently has 100 thousand dollars in their retirement account and intends to take out 50 thousand present day dollars every year after they turn 65 through the age of 80. We show

that to attain a 73.5% probability of maintaining this income stream would require the investor to make annual inflation-adjusted infusions of 20 thousand dollars a year until retiring at age 65. (iv) Finally, we compare our GBWM optimal strategy to the performance of target date funds (TDFs), by considering a TDF with a typical glide path for three index funds representing total domestic bond, total international stock, and total domestic stock. Because our GBWM strategy is on the efficient frontier, uses a wealth dependent strategy, and accounts for investor-specific goals, we will see that our GBWM strategy, which uses the same three index funds, shows a much higher probability of reaching an investor's goals. For example, we will show that using our TDF in the case presented in the previous point reduces the probability of maintaining the desired income stream from 73.5% down to 45.0%.

In the sections that follow, we describe the algorithm (Section 2), expansions of our algorithm to other cases (Section 3), numerical results (Section 4), and our conclusions (Section 5).

2 Algorithm

2.1 Purpose/Intro

We consider a portfolio initially worth $W(0) = W_{init}$ that has access to n different equity assets over the timeframe $0 \leq t \leq T$. Our goal is to dynamically allocate the portfolio among these n assets so that we maximize the probability at $t = T$ of attaining a final portfolio worth at least $W(T) = G$, our goal wealth. We allow for known cash flows $C(t)$ of capital into the portfolio each year. When $C(t) > 0$, we have an infusion into the portfolio. When $C(t) < 0$, we have a withdrawal from the portfolio. All of these cash flows are assumed to be determined at $t = 0$, so they are pre-committed by the investor.

2.2 The Efficient Frontier for Stock Portfolios

For any given portfolio expected return, μ , it is always optimal to minimize the portfolio volatility, σ . Modern portfolio theory, which was developed by Markowitz, see [Markowitz \(1952\)](#), gives a method for the exact allocation among the n assets that gives the minimum σ . If, for every value of μ , we plot the point (σ, μ) where σ is the minimum portfolio volatility from modern portfolio theory, we sketch out a hyperbola in the (σ, μ) plane, which is called the efficient frontier. It is always optimal to maintain the portfolio on the efficient frontier. The question that remains is how to optimally adjust ourselves along the efficient frontier over time.

As shown in [Das, Ostrov, Radhakrishnan, and Srivata \(2018\)](#), the specific hyperbola for the efficient frontier is the equation

$$\sigma = \sqrt{a\mu^2 + b\mu + c}. \quad (1)$$

The constants, a , b , and c are defined by $\mathbf{m}$, which is a vector of the n expected returns; $\mathbf{o}$, which is a vector of n ones; and Σ , which is the $n \times n$ covariance matrix of the n assets, via the following equations:

$$\begin{aligned} a &= \mathbf{h}^\top \Sigma \mathbf{h} \\ b &= 2\mathbf{g}^\top \Sigma \mathbf{h} \\ c &= \mathbf{g}^\top \Sigma \mathbf{g}, \end{aligned}$$

where the vectors $\mathbf{g}$ and $\mathbf{h}$ are defined by

$$\begin{aligned} \mathbf{g} &= \frac{l\Sigma^{-1}\mathbf{o} - \mathbf{k}\Sigma^{-1}\mathbf{m}}{lp - k^2} \\ \mathbf{h} &= \frac{p\Sigma^{-1}\mathbf{m} - \mathbf{k}\Sigma^{-1}\mathbf{o}}{lp - k^2}, \end{aligned}$$

and the scalars k , l , and p are defined by

$$\begin{aligned} k &= \mathbf{m}^\top \Sigma^{-1} \mathbf{o} \\ l &= \mathbf{m}^\top \Sigma^{-1} \mathbf{m} \\ p &= \mathbf{o}^\top \Sigma^{-1} \mathbf{o}. \end{aligned}$$

We allow for restrictions on the segment of the efficient frontier available to the investor, due, for example, to restrictions such as disallowing short positions. We will consider this truncated efficient frontier as the set of potential portfolios from which we optimize the probability of attaining our goal wealth G . More specifically, we define $\mu_{\min}$ and $\mu_{\max}$ to be the smallest and largest values of μ in this truncated efficient frontier. The optimal policy or “control” in our problem will be a value of $\mu \in [\mu_{\min}, \mu_{\max}]$, where the corresponding value of σ , the volatility, is given by equation (1).

2.3 The State Space Gridpoints

The state space consists of time values, t , and wealth values, $W(t)$, which are discretized, so that we consider annual rebalancing (and, later, non-annual rebalancing in Subsection 3.1) at times $t = 0, 1, 2, \dots, T$ and, at each of these times, we use wealth grid points, $W_i(t)$, where the index $i \in \{-i_{\max}(t), -i_{\max}(t)+1, \dots, -1, 0, 1, \dots, i_{\max}(t)-1, i_{\max}(t)\}$. Since we know W_{init} , the initial wealth at $t = 0$, we have that $i_{\max}(0) = 0$ and $W_0(0) = W_{init}$.

To move forward in time, we require a stochastic model for the evolution of the portfolio wealth. We have chosen to use geometric Brownian motion for this paper. That is,

$$W(t) = W(0)e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma\sqrt{t}Z}, \quad (2)$$

where Z is a standard normal random variable, however, we could have just as easily worked with other stochastic models.

Since we are doing annual updates, we first look at $t = 1$. Since the probability that $|Z| > 3.5$ is so small, we will consider a grid whose smallest value corresponds to the value of $W(1)$ in equation (2) if we set $t = 1$, $Z = -3.5$, $W(0) = W_{init}$, $\mu = \mu_{min}$, and σ equal to the largest value it can take, which is the value of σ when $\mu = \mu_{max}$ in equation (1). After we set this smallest value for $W(1)$ equal to $W_{-i_{max}(1)}(1)$, we then compute the largest value of $W(1)$, and set it equal to $W_{i_{max}(1)}(1)$, by plugging into equation (2) the values $t = 1$, $Z = 3.5$, $W(0) = W_{init}$, $\mu = \mu_{max}$, and σ , as before, equal to the largest value it can take.

Since we will choose $i_{max}(t) = t \cdot i_{max}(1)$, the value of $i_{max}(1)$ is important. If it is too big, the dynamic programming algorithm will become slow. If it's too small, the algorithm will lose accuracy. In practice, we have found that $i_{max}(1) = 25$ generally retains sufficient accuracy and speed, as we discuss further in Subsection 4.2.2. The intermediate grid points, where $-i_{max}(1) < i < i_{max}(1)$, are determined from interpolation by satisfying the equation

$$\frac{\ln(W_i(1)) - \ln(W_{-i_{max}(1)})}{\ln(W_{i_{max}(1)}) - \ln(W_{-i_{max}(1)})} = \frac{i - (-i_{max}(1))}{2i_{max}(1)}. \quad (3)$$

We use the logarithm of the wealth here, instead of the wealth, because equation (2) indicates that changes in Z correspond to proportional changes in the logarithm of the wealth.

For $t = 2, 3, \dots, T$ we follow a similar procedure, however, we now have cash flows to consider, so, assuming that $W(t-1) + C(t-1) > 0$, the geometric Brownian motion model now becomes

$$W(t) = (W(t-1) + C(t-1))e^{\left(\mu - \frac{\sigma^2}{2}\right) + \sigma Z}. \quad (4)$$

For the moment, we assume that $W_{-i_{max}(t-1)}(t-1) + C(t-1) > 0$, but in Section 3.2 we will explore the case where the cash flows, $C(t)$, are sufficiently negative that this condition is violated and the investor may become bankrupt. To obtain $W_{-i_{max}(t)}$, we determine $W(t)$ in equation (4) after setting $W(t-1) = W_{-i_{max}(t-1)}$, $Z = -3.5$, $\mu = \mu_{min}$, and σ equal to the largest value it can take. To obtain $W_{i_{max}(t)}$, we set $W(t-1) = W_{i_{max}(t-1)}$, $Z = 3.5$, $\mu = \mu_{max}$, and σ equal to the largest value it can take. The intermediate grid points are then determined by the same interpolation procedure used in equation (3).

2.4 Dynamic Programming for Optimizing the Chance of Obtaining the Investor's Goal

The value function, $V(W(t))$, is the probability that the investor will attain their goal wealth, G , or more at the time horizon T , given they have a worth $W(t)$ at time t . This means that at time T ,

$$V(W_i(T)) = \begin{cases} 0 & \text{if } W_i(T) < G \\ 1 & \text{if } W_i(T) \geq G. \end{cases} \quad (5)$$

We next determine the Bellman equation so that we can determine V at year $t = T - 1$, then $t = T - 2$, etc., iterating backwards in time until we finish at $t = 0$. We begin by determining the transition probabilities, $p(W_j(t+1)|W_i(t), \mu)$. The transition probability is the normalized relative probability that we will be at the wealth node $W_j(t+1)$ at time $t+1$ if we start at the wealth node $W_i(t)$ at time t and, between times t and $t+1$, our portfolio is run with an expected return of μ and its corresponding volatility, σ , from equation (1). Defining $\phi(z)$ to be the value of the probability density function of the standard normal random variable at $Z = z$, we have from equation (4) the following probability density function values

$$\tilde{p}(W_j(t+1)|W_i(t), \mu) = \phi\left(\frac{1}{\sigma}\left(\ln\left(\frac{W_j(t+1)}{W_i(t) + C(t)}\right) - \left(\mu - \frac{\sigma^2}{2}\right)\right)\right), \quad (6)$$

where $C(0) = 0$, since any cash flow at $t = 0$ is incorporated into W_{init} . Normalizing these probability density function values yields the desired transition probabilities:

$$p(W_j(t+1)|W_i(t), \mu) = \frac{\tilde{p}(W_j(t+1)|W_i(t), \mu)}{\sum_{k=-i_{\max}(t+1)}^{i_{\max}(t+1)} \tilde{p}(W_k(t+1)|W_i(t), \mu)}.$$

Since $V(W(t))$ is the expected value of $V(W(T))$, our Bellman equation is simply

$$V(W_i(t)) = \max_{\mu \in [\mu_{\min}, \mu_{\max}]} \left[\sum_{j=-i_{\max}(t+1)}^{i_{\max}(t+1)} V(W_j(t+1)) p(W_j(t+1)|W_i(t), \mu) \right]. \quad (7)$$

We denote $\mu_{i,t}$ as the value of μ at which the maximum is attained in the Bellman equation, and $\sigma_{i,t}$ is, of course, its corresponding volatility on the efficient frontier. As a computational matter, we select an integer m , divide the interval $[\mu_{\min}, \mu_{\max}]$ into an array of m equally spaced values, and let $\mu_{i,t}$ and $V(W_i(t))$ be determined from the μ value within this array that optimizes the sum in the right-hand side of the Bellman equation. In practice $m = 15$ was generally sufficient to maintain accuracy in our results, as we further discuss in Subsection 4.2.1.

First setting $t = T - 1$, we solve the Bellman equation (7) to determine $\mu_{i,T-1}$ and $V(W_i(T - 1))$ for each $i \in [-i_{\max}(T - 1), i_{\max}(T - 1)]$. We then continue backwards in time to $t = T - 2$, $t = T - 3$, etc., until we reach $t = 0$. The value of $V(W_0(0))$ is the optimal probability of the investor attaining their wealth goal G , given their initial wealth $W_0(0) = W_{init}$.

2.5 Probability Distribution for the Investor's Wealth at Future Times

To determine the probability distribution for the investor's wealth at future times, we use the optimal strategy information, $\mu_{i,t}$ and $\sigma_{i,t}$, determined previously from dynamic programming to evolve the probability distribution forward in time, starting with $t = 0$, then $t = 1$, ending at $t = T - 1$. At any given value of t , we determine for each $j \in [-i_{\max}(t + 1), i_{\max}(t + 1)]$

$$p(W_j(t + 1)) = \sum_{i=-i_{\max}(t)}^{i_{\max}(t)} p(W_j(t + 1)|W_i(t), \mu_{i,t}) \cdot p(W_i(t)). \quad (8)$$

Starting with $p(W(0)) = 1$, we generate the entire set of probabilities for $t = 1$, i.e., $p(W_j(1))$ for each $j \in [-i_{\max}(1), i_{\max}(1)]$. Then, moving forward in time, we recursively apply equation (8), until we obtain the probability distribution for the wealth nodes in every year of the lifetime of the portfolio.

2.6 Summary

We summarize the flow and meaning of the dynamic procedure from the investor point of view in four broad steps:

1. The investor determines an initial investment wealth, a goal wealth and a time-frame by which they hope to grow their initial investment into their goal wealth. The investor may also specify annual cash flows for the portfolio, i.e., infusions or withdrawals, if desired.
2. Lower and upper bounds on the mean along the efficient frontier are chosen. (Alternatively, these can be specified through lower and upper bounds on the risk, given by the corresponding standard deviations on the frontier.) These bounds, which determine the specific range of the efficient frontier to which we restrict portfolio choice, can depend on the investor's goal wealth and timeframe, as well as the desire to limit downside or increase upside, as we will explore in Subsection 4.3.2.

3. As the markets move up or down through the investment tenure, our program rebalances the portfolio so that the probability of reaching the goal continues to remain the highest.
4. At any given point in time, we keep track of the portfolio's current wealth, the conditional probabilities of transitioning to different wealth levels at future points in time, and also the probability of meeting the goal under the optimal strategy. This is information that enables the investor to keep track of the performance of the portfolio strategy.

3 Expanding our Dynamic Programming Algorithm to Other Cases

3.1 Non-annual Updates

Our main algorithm is written for annual updates. Annual updates particularly make sense in taxable accounts where short-term capital gains rates are levied on gains realized from stocks held less than a year. For other accounts, however, it might make more sense to update the portfolio every h years, where $h \neq 1$. For example, we might choose $h = 0.25$ if we want quarterly updates. In this context, the integer t now becomes the index of the update, so if $h = 0.25$, then $t = 4$ corresponds to the state of the portfolio after one year and, say, $T = 40$ means that we look to see if we have attained our goal, G , in the 40th update at the end of 10 years.

The main change needed to accommodate these cases where $h \neq 1$ is to note that since ht now represents the actual time represented by index t , equation (2) becomes

$$W(t) = W(0)e^{(\mu - \frac{\sigma^2}{2})ht + \sigma\sqrt{ht}Z},$$

where we note that $W(t)$ is the wealth at index t , which is time ht . This means that equation (4) becomes

$$W(t) = (W(t-1) + C(t-1))e^{(\mu - \frac{\sigma^2}{2})h + \sigma\sqrt{h}Z},$$

and equation (6) becomes

$$\tilde{p}(W_j(t+1)|W_i(t), \mu) = \phi\left(\frac{1}{\sigma\sqrt{h}}\left(\ln\left(\frac{W_j(t+1)}{W_i(t) + C(t)}\right) - \left(\mu - \frac{\sigma^2}{2}\right)h\right)\right).$$

Otherwise, our main algorithm is the same.

If we choose progressively smaller values for h , $i_{\max}(1)$ should be scaled so that $i_{\max}(1) \sim O(h)$. This prevents the computation from sprawling in size, especially

since a small change, h , in time only requires a few additional nodes to accommodate the new range of wealth that is reasonably likely to be attained.

3.2 Incorporating Bankruptcy

When investor withdrawals, $C(t) < 0$, are sufficiently negative, they may cause the investor to go bankrupt. That is, we must consider how to alter our algorithm for these bankruptcy cases where $W_i(t) + C(t) \leq 0$.

To do this, for each time t , we define $i_{pos}(t)$ to be the smallest index i such that $W_i(t) + C(t) > 0$. (The notation “pos” reflects the fact that $W_i(t) + C(t)$ is positive.) This means that for each $i < i_{pos}(t)$, we have a state of bankruptcy after the time t withdrawal, since $W_i(t) + C(t) \leq 0$, while for each $i \geq i_{pos}(t)$, the investor still has money after the time t withdrawal. We note that an investor, once bankrupt, cannot attain their goal wealth, therefore $V(W_i(t)) = 0$ for any $i < i_{pos}(t)$.

If $i_{pos}(t)$ fails to exist at any time t , it means that $W_{i_{max}(t)}(t) + C(t) \leq 0$ at this time, so, from the point of view of our algorithm, the investor is guaranteed to be bankrupt by time t . If $i_{pos}(t)$ exists, we only need to make the following adjustments to our algorithm:

- In Subsection 2.3, to obtain $W_{-i_{max}(t)}(t)$, we determine $W(t)$ in equation (4) after setting $W(t-1) = W_{i_{pos}(t-1)}(t-1)$ instead of $W_{-i_{max}(t-1)}(t-1)$.
- In Subsection 2.4, we still determine the transition probabilities, $p(W_j(t+1)|W_i(t), \mu)$, for all j values, but now only for i values where $i \geq i_{pos}(t)$. After that, the Bellman equation (7) is only used to compute $V(W_i(t))$ for $i \geq i_{pos}(t)$. For $i < i_{pos}(t)$, we have that $V(W_i(t)) = 0$, as stated before.
- Finally, in Subsection 2.5, we alter equation (8) for the probability of being at a wealth node so that the summation over i is from $i_{pos}(t)$ to $i_{max}(t)$ instead of from $-i_{max}(t)$ to $i_{max}(t)$.

We note that the probability of going bankrupt due to the withdrawal at a given time

t is $\sum_{i=-i_{max}(t)}^{i_{pos}(t)-1} p(W_i(t))$, while the probability of being bankrupt prior to this time is $1 - \sum_{i=-i_{max}(t)}^{i_{max}(t)} p(W_i(t))$. In particular, since there is no cash flow at the final time T ,

the probability of the investor going broke by time T is $1 - \sum_{i=-i_{max}(T)}^{i_{max}(T)} p(W_i(T))$.

3.3 Multiple weighted goals

An investor may hope to attain a wealth goal, while at the same time also valuing the goal of not falling below a lower wealth threshold. Further, the investor may want to emphasize the relative importance of one of these wealth goals over the other. Mathematically, this corresponds to letting the lower wealth goal value, G_1 , have a weight of w_1 and the higher wealth goal value, G_2 , have a weight of w_2 , where $w_1 + w_2 = 1$ and the higher w_1 is, the more important goal G_1 is relative to goal G_2 . To accommodate these two goals, we simply replace the terminal value equation (5) with

$$V(W_i(T)) = \begin{cases} 0 & \text{if } W_i(T) < G_1 \\ w_1 & \text{if } G_1 \leq W_i(T) < G_2 \\ 1 & \text{if } W_i(T) \geq G_2, \end{cases}$$

and run the algorithm as before, noting that the value function V no longer represents the probability of attaining the goal wealth, since there is no longer a single goal wealth. If desired, we can easily extend this to k wealth goals $G_1 < G_2 < \dots < G_k$ with weights $w_1, w_2, \dots, w_k$ that sum to one, by replacing the terminal value equation (5) with

$$V(W_i(T)) = \begin{cases} 0 & \text{if } W_i(T) < G_1 \\ \sum_{l=1}^j w_l & \text{if } G_j \leq W_i(T) < G_{j+1} \text{ for } j = 1, 2, \dots, k-1 \\ 1 & \text{if } W_i(T) \geq G_k. \end{cases}$$

4 Results

In this section, we begin by describing a base case, which we then fine tune by adjusting our algorithm's parameters so as to optimize the algorithm's speed vs. accuracy trade-offs. We then demonstrate the results of the algorithm for a number of cases, both with and without periodic infusions and withdrawals. At the end of the section we demonstrate how our algorithm can be used to minimize the probability of an investor going bankrupt during retirement, how an investor can understand the effect of periodic infusions on the minimum probability of going bankrupt, and how our algorithm compares to target date fund performance for achieving goals.

4.1 Base case

We create a base case from which we will later compute comparative statics. For our base case, we assume the investor begins with a \$100 investment at $t = 0$. The investor's goal is to maximize their probability of reaching a goal wealth of $G = \$200$

at the end of year $T = 10$. We note that the ratio of these wealth values is all that is important here. That is, the maximum probability of going from a wealth of \$100 to at least \$200 at $T = 10$ in our base case is the same as the maximum probability of doubling any initial investment at $T = 10$.

No cash flows are present in the base case, although we will certainly explore their effect later. The base case value for $i_{\max}(1)$, which specifies the rate of growth in the number of wealth nodes per period, is 25. The base case value for m , the number of potential portfolios we consider along the efficient frontier, is 15. These values for $i_{\max}(1)$ and m have been chosen for reasons that will be explained in Subsection 4.2.

The efficient frontier arising from the investments available to the investor in our base case is exogenously determined. For illustrative purposes, we have generated this efficient frontier using historical returns from the 20 year period between January 1998 to December 2017 for index funds representing US Bonds, International Stocks, and US Stocks³. The mean and covariance of returns for these indexes are given in Table 1.

Table 1: Summary statistics on returns from January 1998 to December 2017 for our three index funds.

Index fund category	Mean Return	Covariance of Returns		
U.S. Bonds	0.0493	0.0017	-0.0017	-0.0021
International Stocks	0.0770	-0.0017	0.0396	0.03086
U.S. Stocks	0.0886	-0.0021	0.0309	0.0392

The data from this table is used in conjunction with the mathematics in Subsection 2.2 to generate the efficient frontier, shown in Figure 1. The range for μ is restricted so that $\mu_{\min} \leq \mu \leq \mu_{\max}$. The bounds, $\mu_{\min}$ and $\mu_{\max}$, can be chosen through a variety of methods, including bounds on the investor's tolerance for risk via the standard deviation on the efficient portfolio or, as we will explore in Subsection 4.3.2, the investor's interest in limiting the downside or increasing the potential upside in the portfolio's final wealth distribution. For our base case, we have selected $\mu_{\min} = 0.0526$ to correspond to the lowest possible portfolio standard deviation on the efficient frontier, which is $\sigma = 0.0374$. We have selected $\mu_{\max} = 0.0886$, the highest mean return of the three index funds, so as to avoid long-short portfolios. This value of $\mu_{\max}$ corresponds to $\sigma = 0.1954$. These numbers are realistic and match those in

³The three index funds used are (i) Vanguard Total Bond Market II Index Fund Investor Shares (VTBIX), representative of U.S. Fixed Income (Intermediate-Term Bond), (ii) Vanguard Total International Stock Index Fund Investor Shares (VGTIX), representative of Global Equity (Large Cap Blend), (iii) Vanguard Total Stock Market Index Fund Investor Shares (VTSMX), representative of U.S. Equity (Large Cap Blend). These three funds have been chosen only as representatives of their respective asset categories for illustrative purposes.

related research, see for example Exhibit 5 in Wang, Suri, Laster, Almadi (2011). We consider $m = 15$ portfolios on the efficient frontier whose μ values are equally spaced over the interval $[\mu_{\min}, \mu_{\max}]$. At each year and wealth node in the state space, the dynamic strategy determines which of these 15 portfolios is optimal.

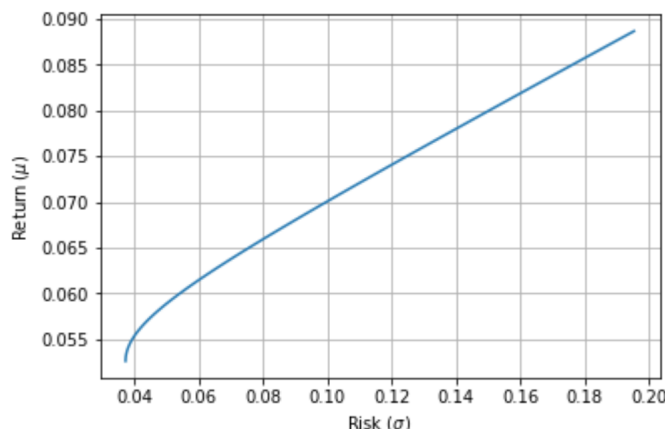


Figure 1: Efficient frontier generated from data on the returns of our three indexes, shown in Table 1.

Under the optimal dynamic strategy, we find that the highest achievable probability of reaching the goal wealth or more is $Pr[W(T) \geq G = 200] = 0.666$. The initial portfolio has $\mu = 0.0835$ and $\sigma = 0.1686$. Figure 2 shows the cumulative distribution of wealth for all $t = 1, 2, \dots, 10$ years. The evolution of this distribution shows the shift of these wealth distributions to the right as time proceeds, but, more interestingly, it also shows the adjustments the distribution shape makes so as to maximize the probability of exceeding the value of $G = 200$ by the final year $T = 10$. In the earlier years, the distribution has a slight positive skew, as is the case for a lognormal distribution, but the adjustments to attain the goal wealth eventually reverse this and create a negative skew to the distribution as it progresses to its time horizon.

Figure 3 shows the probability of reaching our goal wealth $G = 200$ (i.e., the value function) at each point in time and for any level of wealth in the state space; that is, at each $\{t, W_i(t)\}$ grid point. As is expected, higher wealth levels are associated with higher probabilities of reaching the goal wealth. The figure also reflects the fact that as we reach the final time $T = 10$, we have more certainty about whether we will attain the goal wealth.

In Figure 4, we show the optimal portfolio strategy at each $\{t, W_i(t)\}$ grid point. That is, we show which of the 15 μ values are optimal, where portfolio number 0 corresponds to $\mu_{\min}$ and portfolio number 14 corresponds to $\mu_{\max}$. When the portfolio has a lot of money, it moves towards lower portfolio numbers, since the corresponding decrease in volatility makes it less likely to incur big losses that could remove investors

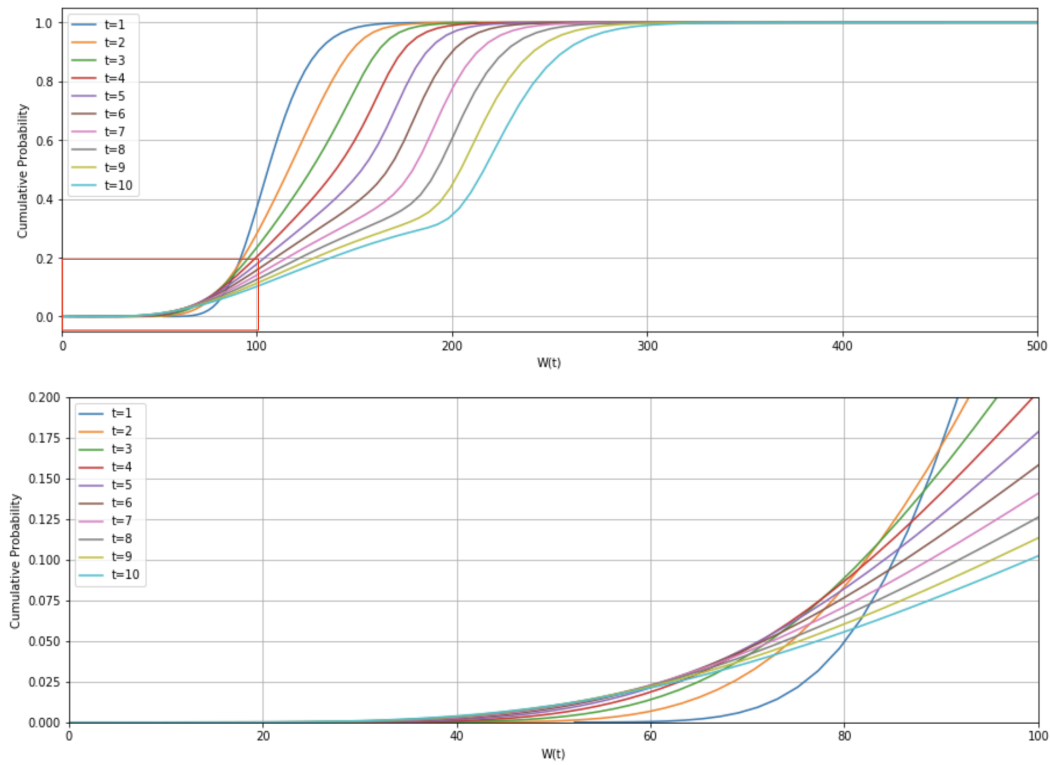


Figure 2: Annual evolution of the cumulative probability function for wealth under the optimal dynamic strategy. The second plot zooms in on the bottom left square of the upper plot (outlined in red), better showing the distributions' left tails.

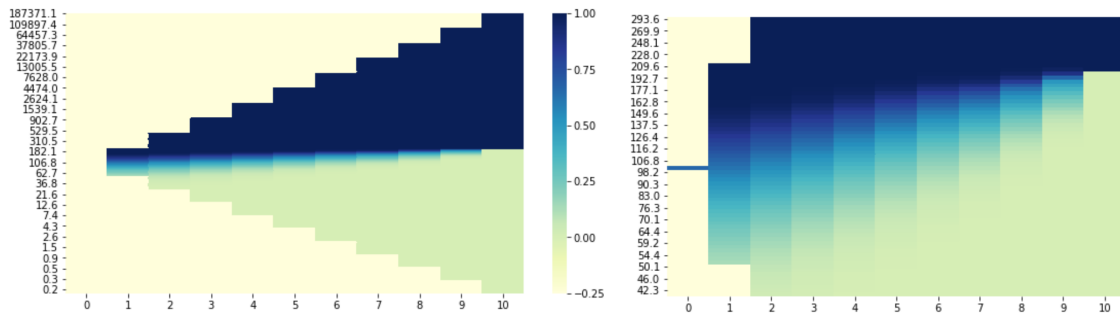


Figure 3: Probability of reaching the goal wealth $G = 200$ at each wealth node and time under the optimal dynamic strategy. The x -axis shows the time in years and the y -axis depicts the level of wealth. Note that the y -axis is an exponentially increasing scale. The left panel shows the entire figure, and the right panel is zoomed in on the range of points near the center of the left panel where the probability varies.

from the path to attaining the goal wealth that they are currently on. When the portfolio has less money, it moves towards higher portfolio numbers, since the increase

in both expected return and volatility makes it more likely to attain the goal wealth.

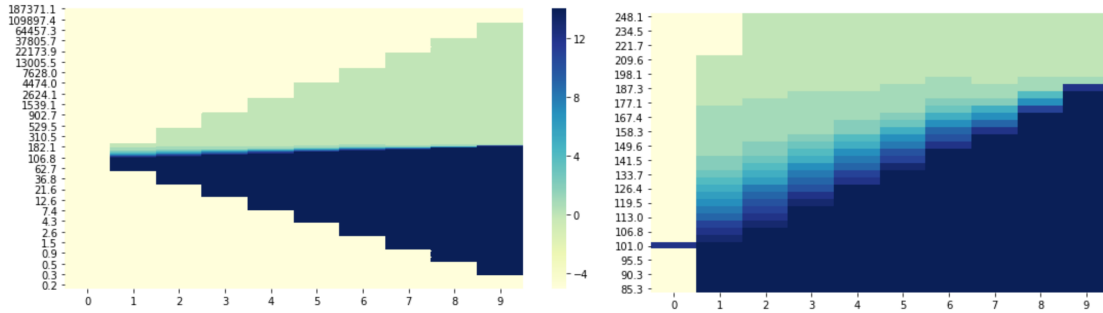


Figure 4: Optimal portfolio strategy at each wealth node and time. Note that portfolio numbers run from 0 to 14 for the $m = 15$ model portfolios on the efficient frontier. The x -axis shows the time in years and the y -axis depicts the level of wealth on an exponentially increasing scale. The left panel shows the entire figure, and the right panel is zoomed in on the range of points near the center of the left panel where portfolio choice varies.

4.2 Fine tuning the parameters to balance algorithm speed vs. accuracy

4.2.1 Effect of changing, m , the number of portfolio strategies

In Table 2, we consider the effect on the base case of changing the number of intermediate strategies we consider on the efficient frontier between $\mu = \mu_{\min}$ and $\mu = \mu_{\max}$. Because the efficient frontier is part of a hyperbola, it becomes progressively linear as μ increases. This means the intermediate strategies matter more near $\mu_{\min}$. From Table 2, we see that $m = 15$, our base case value (denoted by an asterisk in the table), is more than sufficient for providing enough accuracy in determining the probability of the initial investment $W_{init} = 100$ gaining at least 50% and at least 100% of its initial worth after 10 years. The table also shows that the rate of growth in the run time as m increases is approximately linear.

4.2.2 Effect of changing the number of grid points, $i_{\max}(1)$

In Table 3, we consider the effect on the base case of changing the rate at which the wealth grid points increase with each time period. Recall that at $t = 0$, we have only one grid point corresponding to $W(0) = 100$, and then we add $2 \cdot i_{\max}(1)$ grid points with each time period, so we end at time $T = 10$ with $20 \cdot i_{\max}(1) + 1$ grid points. We note from Table 3 that the rate of growth in the run time as $i_{\max}(1)$ increases is approximately quadratic. The probabilities of the initial investment gaining at least

Table 2: Effect of changing m , the number of strategies

value of m	run time (sec.)	$Pr[W(T) \geq 150]$	$Pr[W(T) \geq G = 200]$
5	.763	.765	.663
10	.959	.772	.666
15*	1.21	.772	.666
20	1.41	.773	.667
40	2.30	.772	.667
60	3.15	.772	.667
80	3.94	.772	.667
100	4.88	.772	.667

50% and at least 100% of its initial worth have some noise that dies down rather slowly as $i_{\max}(1)$ increases. We choose $i_{\max}(1) = 25$ for the base case (again, denoted with an asterisk) since the noise is within reason while the run time hasn't grown too large.

Table 3: Effect of changing $i_{\max}(1)$, the additional grid points per period

value of $i_{\max}(1)$	run time (sec.)	$Pr[W(T) \geq 150]$	$Pr[W(T) \geq G = 200]$
10	0.37	.781	.680
15	0.61	.787	.685
20	0.90	.775	.657
25*	1.21	.772	.666
50	3.71	.777	.672
75	7.75	.779	.664
100	12.40	.777	.669
150	25.10	.778	.672

4.3 The effect of the investor making changes

4.3.1 Effect of changing T , the portfolio's time horizon

In Figure 4, we look at the effect of changing the time horizon T for the portfolio. In this figure, we let the goal wealth G increase linearly with T instead of exponentially, as G actually scales with T . As a result, we see the associated probability of attaining G grow as T increases in the table.

The number of wealth nodes grows linearly as we increase t . In particular, at the final time $t = T$, there are $2T \cdot i_{\max}(1) + 1$ grid points, and the total number of grid points used at all times is $(T + 1)(T \cdot i_{\max}(1) + 1)$. The additional grid points created when we double T appear to increase the run time by a factor of approximately six.

That is, the run time appears to grow at a rate that is between quadratic and cubic in T , which correspond to increased run time factors of four and eight respectively.

Table 4: Effect of changing T , the portfolio's time horizon

time horizon T	run time (sec.)	Goal wealth, G	$Pr[W(T) \geq G]$
5	0.21	150	.567
10*	1.21	200*	.666
15	3.34	250	.756
20	7.03	300	.846
25	12.9	350	.918
30	21.2	400	.958
40	46.3	500	.995

4.3.2 Effect of changing $\mu_{\min}$ or $\mu_{\max}$ for the efficient frontier truncation

Our algorithm can accommodate any given set of funds from which it then forms allowable portfolios along the efficient frontier. This has two benefits. First, because the funds are selected independently from the mechanics of the algorithm, the determination of the funds and the efficient frontier can be determined by a different operating team in the fund management business from the team running the dynamic programming algorithm. Further, if different sectors of the fund management business need to work with different funds, our algorithm can easily accommodate each sector separately. Second, the spread in the wealth distribution at time T can be controlled to some degree by changing the endpoints of the interval $\mu \in [\mu_{\min}, \mu_{\max}]$ that restrict the (σ, μ) pairs on the efficient frontier available to our algorithm. In this subsection we explore this second benefit by altering $\mu_{\min}$ and then $\mu_{\max}$ in our base case.

Recall that in our base case we consider 15 (σ, μ) pairs on the efficient frontier, with the lower end of the frontier at $(0.0374, 0.0526)$ and the upper end at $(0.1954, 0.0886)$, so $\mu_{\min} = 0.0526$ and $\mu_{\max} = 0.0886$. In the top panel of Figure 5, we see the effect on the terminal distribution of wealth when we chose three different values for $\mu_{\min}$: 0.0526, 0.06, and 0.07. As $\mu_{\min}$ increases, the probability of attaining the goal wealth $G = 200$ goes down since the interval of available controls shrinks. Also, the wealth distribution has a higher variance, as is to be expected. But these higher risk distributions also have higher positive skewness, evident from the longer right tails. Therefore, choosing the value of $\mu_{\min}$ corresponds to choosing a trade-off between variance and skewness. More notably, the left tails of all three distributions are very similar, indicating that modulating $\mu_{\min}$ has a much stronger effect on the right side of the wealth distribution.

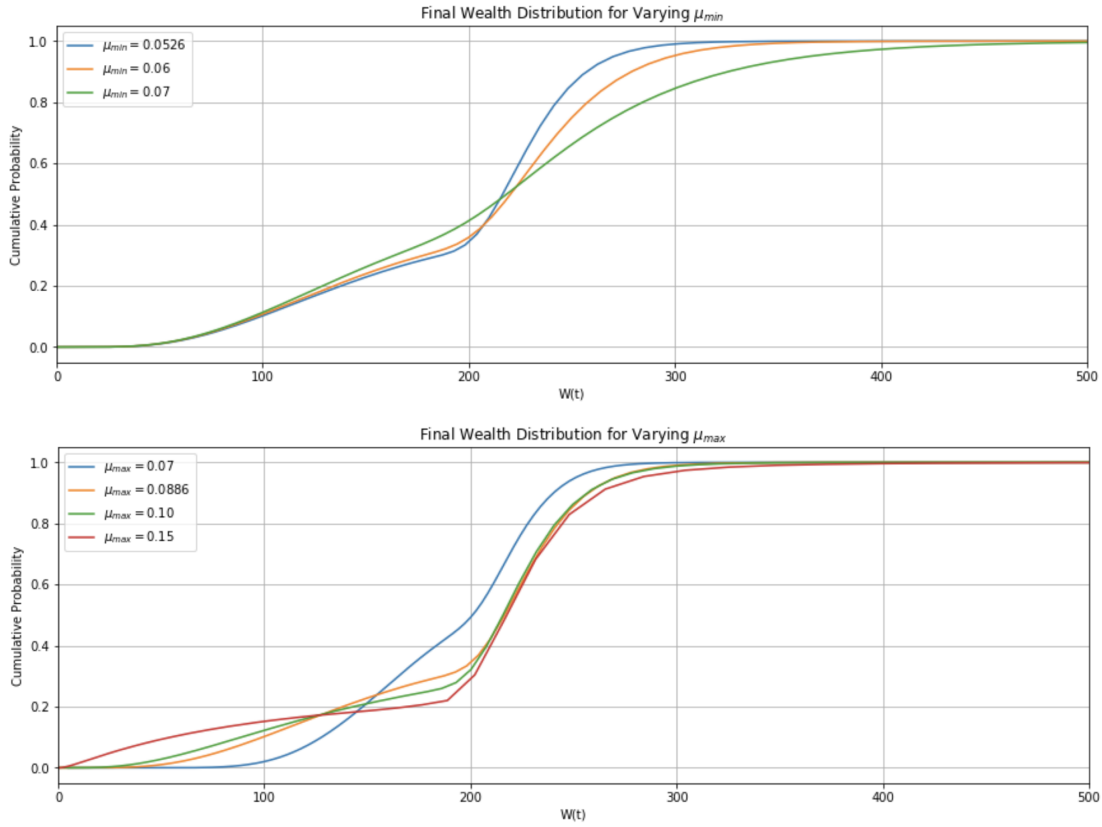


Figure 5: Terminal wealth cumulative probability distributions when μ_{min} and μ_{max} are varied. All other parameters from the base case remain the same. The ranges are: $\mu_{min} = \{0.0526, 0.06, 0.07\}$ and $\mu_{max} = \{0.07, 0.0886, 0.10, 0.15\}$.

The lower panel of Figure 5 shows the effect of varying μ_{max} by considering four different μ_{max} values: 0.07, 0.0886, 0.10, and 0.15. Again, the probability of attaining the goal wealth $G = 200$ increases as μ_{max} increases, because the interval of available controls grows. However, in this case we notice that both the left and right sides of the probability distribution are affected, although the effect on the downside is more pronounced than that on the upside, suggesting that varying μ_{max} has a greater effect on the left tail.

An investor that is more accepting of risk would tend to first want to increase μ_{max} , but Figure 5 suggests that increasing μ_{min} might be the wiser course of action, since we can see in this case that increasing μ_{min} appears to have a stronger influence on the upside potential whereas increasing μ_{max} appears to have a stronger influence on the downside, risking significant losses without that much compensating gains.

The reason for this is actually straightforward: The algorithm is only interested in attaining the goal wealth, so the optimal strategy for a well-off investor is to move

μ to μ_{min} so as to reduce volatility and the chance of major losses resulting in no longer being on track to attain the goal wealth. Because the well-off investor uses μ_{min} , increasing μ_{max} has little effect on the right tail, while increasing μ_{min} has a significant effect. Similarly, when the investor is worse off, they select μ_{max} because that increases both μ and σ , which increase the chance of big gains and attaining the goal wealth. Of course this also increases the chance of big losses, which inflates the left tail of the wealth distribution.

4.3.3 The effect of cash flows: infusions or withdrawals

1. Annual infusions: $C(t) > 0$

We continue to work with our base case where we have an initial investment of $W(0) = W_{init} = 100$ thousand dollars and a goal of having at least $W(T) = G = 200$ thousand dollars at the end of $T = 10$ years. In Table 5, we look at how constant annual infusions of $C(t) = 1, 2, \dots, 9$ thousand dollars affect the maximum probability of reaching this goal, as well as the probability of reaching at least 150 thousand dollars. We note from the table that even small infusions can have a significant effect on increasing these probabilities.

Table 5: Effect of changing annual infusions, $C(t) \geq 0$

value of $C(t)$	$Pr[W(T) \geq 150]$	$Pr[W(T) \geq G = 200]$
0*	.772	.666
1	.830	.720
2	.883	.798
3	.924	.839
4	.963	.906
5	.987	.953
6	.995	.973
7	.999	.990
8	.999	.998
9	.999	.999

2. Annual withdrawals: $C(t) < 0$ and the probability of going bankrupt.

We now look at the same situation, but with constant annual withdrawals instead of infusions, so $C(t)$ is now a negative constant. Should the annual withdrawal amount become significant, the investor will now risk bankruptcy (i.e., $W(T) = 0$). In Table 6, we see how increasing the withdrawal rate increases the chance of the investor going bankrupt, while decreasing both the probability of reaching 150 thousand dollars and the probability of reaching the goal wealth of 200 thousand dollars at time $T = 10$.

Table 6: Effect of changing annual withdrawals, $C(t) \leq 0$

value of $C(t)$	$Pr[W(T) = 0]$	$Pr[W(T) \geq 150]$	$Pr[W(T) \geq G = 200]$
0*	0	.772	.666
-1	0	.718	.617
-5	.001	.490	.380
-10	.118	.247	.189
-15	.483	.098	.073
-20	.793	.035	.024
-25	.935	.011	.007

4.3.4 Attaining retirement goals: our algorithm versus a target date approach

Our algorithm can be used to solve a variety of important problems for optimizing retirement savings. Here, for example, we consider the effect of infusions on the maximized probability of not running out of money in retirement. In particular, we consider at $t = 0$ a 50 year old investor who currently has 100 thousand dollars in their retirement account and intends to take out 50 thousand present day dollars every year after they turn 65 through the age of 80, where $t = T = 30$. We assume that the annual rate of inflation is 3%, so, in thousands of dollars, that means that at age 66, $C(t) = -50 \cdot 1.03^{16}$, at age 67, $C(t) = -50 \cdot 1.03^{17}$, up through age 79 where $C(t) = -50 \cdot 1.03^{29}$. Because $C(t)$ isn't defined at time T , which corresponds to when the investor is 80, we need to set the goal wealth G equal, in thousands of dollars, to $G = 50 \cdot 1.03^{30} = 121.4$, so that the investor can make their final withdrawal without going bankrupt.

Our algorithm now optimizes the chance that the investor does not go bankrupt, but it finds, unfortunately, that this optimal probability is only 12.8%. The investor, therefore, considers making infusions of c thousand present day dollars each year before retiring at age 65, starting with an infusion of $c \cdot 1.03$ at the age of 51. The effect of increasing c on the maximized probability of remaining solvent at age 80 is given in the first two columns of Table 7.

Target Date Funds (TDFs) play an important role in the space of retirement investing. They provide a logical investment strategy that has the advantage of being customized to the age of an investor. Our GBWM approach allows the investment strategy to be further customized to the investor's needs by considering their goals, timeframe, cash flows, and state of wealth over time, in addition to their age.

To quantify the advantages this additional customization provides, we have, for comparison to our GBWM results, created a hypothetical TDF using the same three index funds representative of U.S. Bonds, International Stocks, and U.S. Stocks used

Table 7: Effect of changing pre-retirement infusions, c , for our algorithm with the goal of staying solvent and for our Target Date Fund

Value of c	Our GBWM algorithm: probability of staying solvent	Our Target Date Fund: probability of staying solvent
0	12.8%	0.7%
5	25.8%	3.9%
10	42.1%	11.9%
15	58.6%	26.6%
20	73.5%	45.0%
25	85.4%	62.7%
30	93.8%	77.0%
35	98.4%	86.6%
40	99.8%	92.8%

by our GBWM algorithm, along with a typical glide path, shown in Table 8, for an investor transitioning from age 50 to age 80. Using the allocations from the glide path in this table, we use simulation to determine the probabilities of an investor remaining solvent at age 80. This gives us the results shown in the Target Date Fund column in Table 7.

Table 8: Our Target Date Fund glide path

Age range	50–54	55–59	60–64	65–69	70–74	75–80
1. U.S. Stock	44%	40%	35%	28%	20%	18%
2. International Stock	29%	26%	23%	19%	13%	12%
3. U.S. Bond	27%	34%	42%	53%	67%	70%

We see from Table 7 that our algorithm shows a significantly higher ability to achieve the investor’s goal of staying solvent. More specifically, the advantage in using our GBWM algorithm is greater than 30 percentage points when $c = 10$ or 15 and remains high for the other values of c as well.

There are a number of reasons for our algorithm’s considerable outperformance. One reason is that the TDF retirement allocations are not generally on the efficient frontier, so they incur some additional volatility that is not compensated by increased expected returns. A second reason is that the allocation within TDFs is time dependent, but does not depend on the level of the investor’s wealth, whereas our algorithm’s allocation strategy depends on both time and the level of wealth. Finally, our algorithm takes into account the investor’s stated goals and decisions, specifically the infusions and the withdrawals the investor has pre-determined, as well as the investor’s timeframe and the goal of staying solvent at the end of this timeframe.

By allowing our optimization to be customizable to an individual investor's circumstances, specifications, and goals, our algorithm gains a considerable advantage over the one-size-fits-all nature of target date funds. Table 7 shows that these differences are not small.

We can better understand the trade-offs between our GBWM algorithm and our Target Date Fund by comparing the two panels in Figure 6, where we present the cumulative probability distributions at $t = \{5, 10, 15, 20, 25, 30\}$ years for our GBWM methodology and our Target Date Fund. For this figure we have chosen $c = 25$, which is close to the current 24 thousand dollar annual limit on 401k contributions for employees who are 50 or older. We note that the right tail of our Target Date Fund at $t = 30$ is higher. We also note at the times earlier than $t = 30$ that are displayed, the chance of going bankrupt, which is 1 minus the value of the graph at $W = 0$, is slightly higher for our GBWM methodology than our TDF. These are the trade-offs that lead to our GBWM methodology attaining a much higher probability of not going bankrupt at $t = 30$.

In Table 9, we change the goal of our algorithm from staying solvent to ending with a balance at or above \$500,000. This has no effect on our Target Date Fund numbers, of course, but for our algorithm, as expected, it lowers the probability of staying solvent, while increasing the probability of ending above \$500,000. Again, we see that the advantage in using our GBWM methodology is greater than 30 percentage points, this time when $c = 15, 20$, or 25, and, again, it remains high for the other values of c as well. The effect of changing the GBWM goal to having a balance at or above \$500,000 on the cumulative distribution over time can be seen in the top panel of Figure 7. Note in this panel the evolution of the advantageous kink that finally lands a little after \$500 thousand in the $t = T = 30$ curve.

Finally, in Table 10, we show the results of dividing our goal, as discussed in Subsection 3.3, between staying solvent with a 60% weight and ending with a balance at or above \$500,000 with a 40% weight. Comparing the results in Table 10 with the results in Table 7, we see that having a 60% weight, as opposed to the full weight, on the goal of staying solvent leads to losses in the probability for attaining this goal of less than 8 percentage points. Even better, comparing Table 10 with Table 9 for the goal of attaining at least \$500,000, we only observe losses of less than 1 percentage point. The evolution of the cumulative distribution for this mixed GBWM goal when $c = 25$ can be seen in the bottom panel of Figure 7.

5 Concluding Discussion

In this paper, we have developed an algorithm that can quickly determine (generally within a few seconds to a minute) an optimal dynamic trading strategy for goals-based

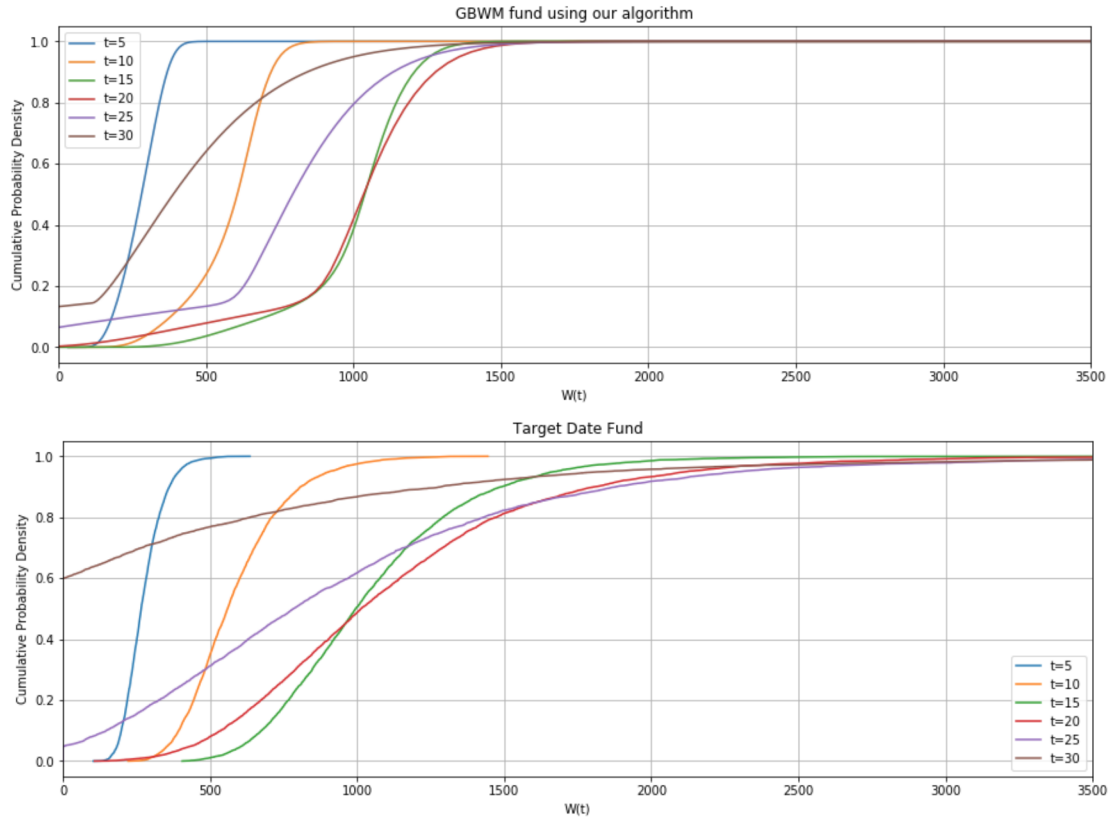


Figure 6: Cumulative probability distributions when c is 25 for the wealth as it evolves across $t = \{5, 10, 15, 20, 25, 30\}$ years. The top panel depicts the cumulative distribution when the goal of our GBWM methodology is to remain solvent at $T = 30$. The bottom panel shows the cumulative wealth distribution for our Target Date Fund.

Table 9: Effect of changing pre-retirement infusions, c , for our algorithm with the goal of ending with at least \$500,000 and for our Target Date Fund

Value of c	Our algorithm: probability of ending with $\geq \$500,000$	Our Target Date Fund: probability of ending with $\geq \$500,000$
0	8.9%	0.2%
5	18.5%	1.2%
10	31.2%	4.2%
15	45.1%	12.0%
20	58.6%	24.5%
25	70.9%	39.9%
30	81.3%	55.6%
35	89.1%	69.5%
40	94.9%	80.1%

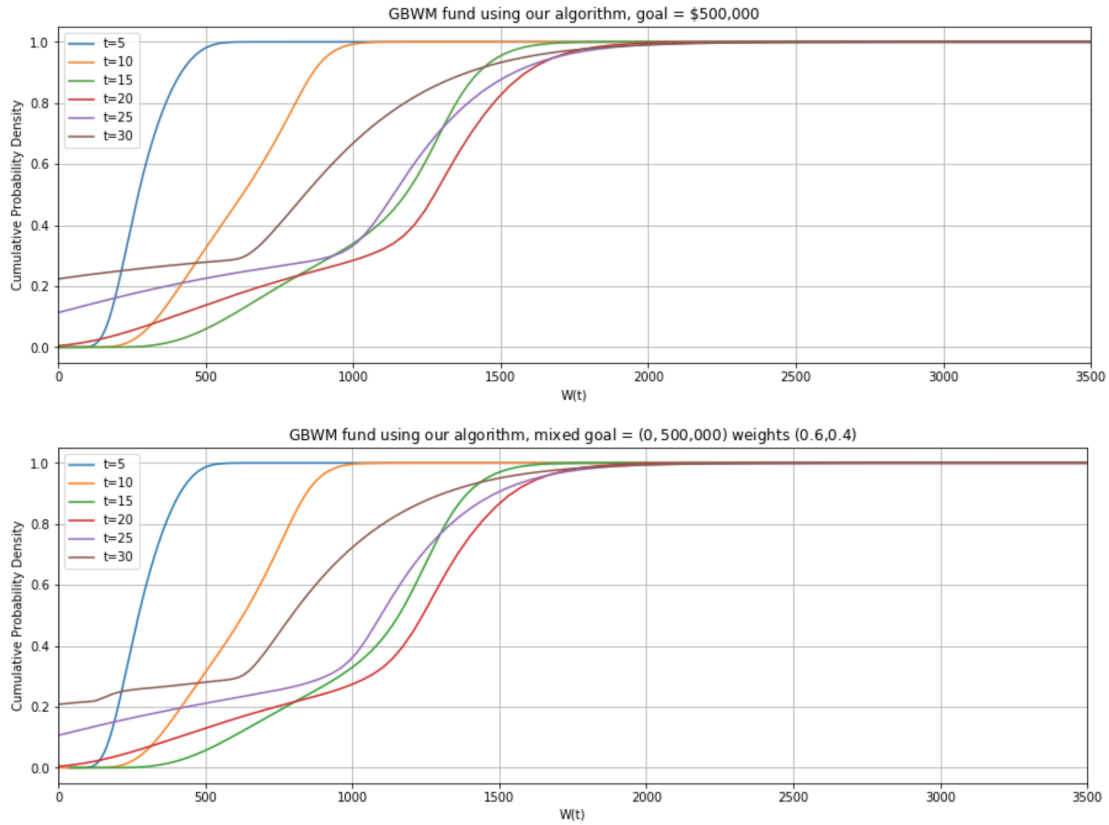


Figure 7: Cumulative probability distributions when c is 25 for the wealth as it evolves across $t = \{5, 10, 15, 20, 25, 30\}$ years. The top panel depicts the cumulative distribution when the goal of our GBWM methodology is to deliver an amount greater than \$500,000 at $T = 30$. The bottom panel displays the cumulative distribution when we split the goal for our GBWM methodology between staying solvent with a 60% weight and delivering an amount greater than \$500,000 with a 40% weight at $T = 30$.

wealth management (GBWM). The objective function we maximize in our GBWM framework is the probability of reaching a given goal wealth at the end of a given investment horizon, in contrast to maximizing the expected utility of the wealth at the end of a given time horizon. Without any sacrifice in runtime, our algorithm can optimally allocate from among any given set, large or small, of chosen portfolios, which can be determined outside of our algorithm. Also, without any sacrifice in runtime, our algorithm can accommodate periodic specified infusions or withdrawals. Further, we can easily compute the probability of running out of money at each period due to any of these withdrawals.

The dynamic programming approach, whose computation works backwards in time, has important advantages over approaches that work forwards in time. Dynamic programming addresses the fact that the optimal portfolio allocation will shift

Table 10: Effect of changing pre-retirement infusions, c , for our algorithm with the split goal of staying solvent with a 60% weight and ending with at least \$500,000 with a 40% weight.

Value of c	Our algorithm: probability of staying solvent	Our algorithm: probability of ending with $\geq$ \$500,000
0	11.1%	8.7%
5	22.6%	18.2%
10	37.1%	30.8%
15	52.3%	44.8%
20	66.2%	58.0%
25	78.2%	70.3%
30	87.5%	80.8%
35	94.1%	88.7%
40	98.0%	94.7%

over time. Attempting to consider this with a forwards in time algorithm is computationally infeasible, so these algorithms are reduced to myopic approaches, where the allocation chosen at a point in time is only optimal under the assumption that no further reallocation will occur. That is, the forwards in time algorithms are restricted to the inferior approach of repeated static optimization. Interestingly, despite this, the results from the repeated static optimization method described in [Das, Ostrov, Radhakrishnan, and Srivata](#) (2018) end up being very similar to the results obtained from our dynamic programming approach, however, that paper’s method, which relies on the solution to a stochastic differential equation (SDE), cannot be extended to infusions or withdrawals and it has a longer computational time. Most forwards in time approaches rely on simulation instead of solving an SDE, which leads to even longer computational times to be reasonably accurate.

We can modulate the left and right tails of the terminal wealth distribution by varying the chosen range of allowable portfolios on the efficient frontier. If we remove either the most risky investments at the top of this range or the least risky investments at the bottom of this range, we lower the probability of reaching our goal wealth, since we are restricting the set of controls available to the algorithm. However, removing the most risky investments has the benefit of lowering the left tail in the distribution, as it lowers the chance of significant losses, while removing the least risky investments has the benefit of raising the right tail in the distribution, making it more likely to attain wealth values that are significantly higher than the goal wealth.

Target date (or life-cycle) funds have the important feature of being able to re-allocate funds based on time, specifically on the age of the investor. They cannot, however, reallocate based on the investor’s wealth or the investor’s specific goals. Nor are the allocations underlying a target date fund necessarily on the efficient fron-

tier. The natural question arises as to whether incorporating these GBWM features and efficiency constraints, which require individualized attention through automation and/or human advisors, adds much value to the investor. Our paper concludes that they can actually add considerable value. We explored the case of a 50 year old saving for retirement at age 65, who then takes out inflation adjusted withdrawals until the age of 80. Working with just three index funds representative of U.S. Bonds, International Stocks, and U.S. Stocks, we show examples where our algorithm's optimal dynamic reallocation strategy increases the chance of this investor remaining solvent at age 80 by more than 30 percentage points over using a Target Date Fund during this period.

The approach in this paper can be expanded to answer questions arising in other important financial situations. These include incorporating the effect of taxes or incorporating stochastic efficient frontier models. Because our algorithm runs so quickly, it is unlikely that these additional features will degrade the runtime particularly significantly.

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